

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459976.7833 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.089 +/- 0.012
Transit depth (Rp/Rs)^2	0.78 +/- 0.21 [%]
Orbital Inclination (inc)	88.09 +/- 0.27
Airmass coefficient 1 (a1)	1.1141 +/- 0.0058
Airmass coefficient 2 (a2)	-0.0847 +/- 0.0076
Scatter in the residuals of the lightcurve fit is	0.52 %
Best Comparison Star	None
Optimal Aperture	4.6
Optimal Annulus	5.92
Transit Duration (day)	0.099 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.089 $\pm$ 0.012 vs 0.1295 (deviation 3.26 σ)
Duration fit vs published	0.099 d vs 0.1146 d (deviation 1.16 σ)
Mid-time O-C	4.2 min
Depth SNR	7.2
Residual scatter	0.52 %

Light curve

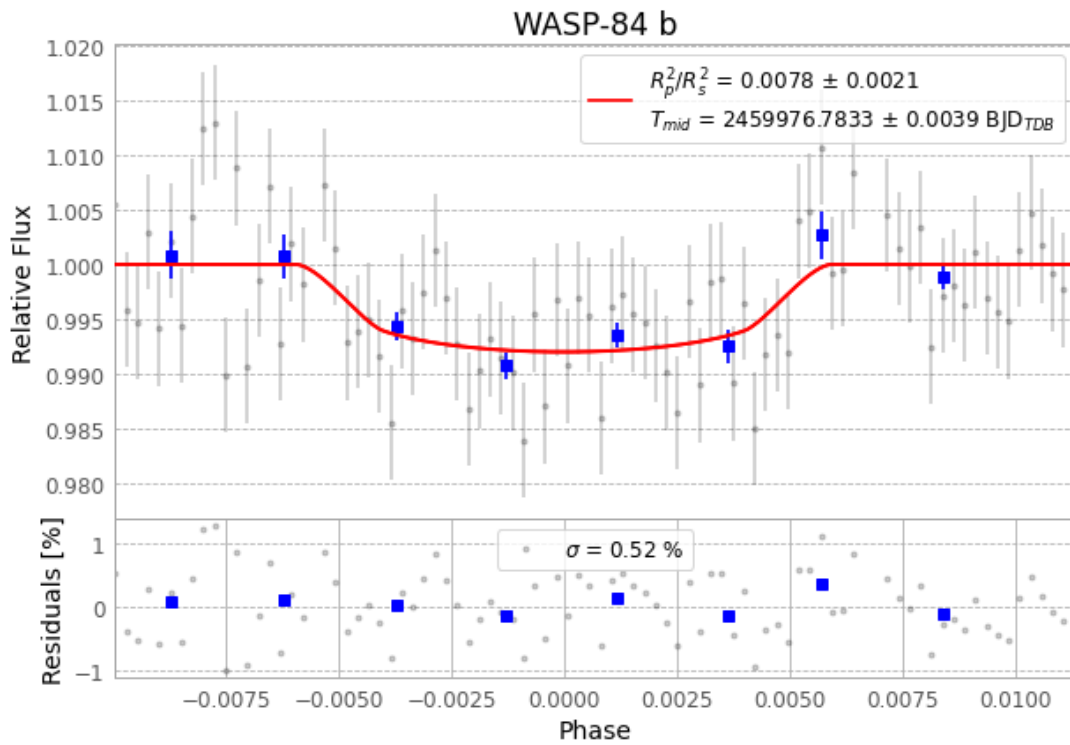


Figure 1 — FinalLightCurve_WASP-84 b_2023-01-31.png (SHA-256 2ec9845449539d65...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 114], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 39c0f76fa35dfe14...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

88 FITS frames (58 MB), WASP-84230201043615.FITS → WASP-84230201085716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-84-230201.log (a5a7feab9980...), FinalLightCurve_WASP-84 b_2023-01-31.png (2ec984544953...), FinalLightCurve_WASP-84 b_2023-01-31.csv (f316a792cc04...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 02:16Z.